



QSKILL • SQUARCELL RESOURCE INDIA PVT. LTD.

Python Development Internship

A summary of the projects, tools, and outcomes from a one-month virtual internship in Python Development.

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What This Covers



Internship Overview

Organization, mode, duration, and objectives



Project 1: Data Analysis

Student score analysis using pandas & matplotlib



Project 2: Matrix Operations Tool

A NumPy-based command-line calculator



Project 3: Linear Regression

House price prediction using scikit-learn



Tools & Technologies

The Python stack used across all three projects



Outcomes & Certificate

Results achieved and proof of completion



Internship Overview

Squarcell Resource India Pvt. Ltd., operating under the brand QSkill, is an ISO 9001:2015 certified organization based in Navi Mumbai, Maharashtra, offering virtual, project-based internship programs.

ROLE

Python Development Intern

MODE

Virtual

DURATION

1 Month (25 Jul – 25 Aug 2026)

TIMING

Flexible

FOCUS

Learning, skill development, hands-on projects

PROJECT 1

Data Analysis Task

Analyzed a dataset of 20 students across 5 subjects using pandas: computed subject-wise summary statistics, derived each student's average score, ranked performance, and visualized results with matplotlib.

20

Students Analyzed

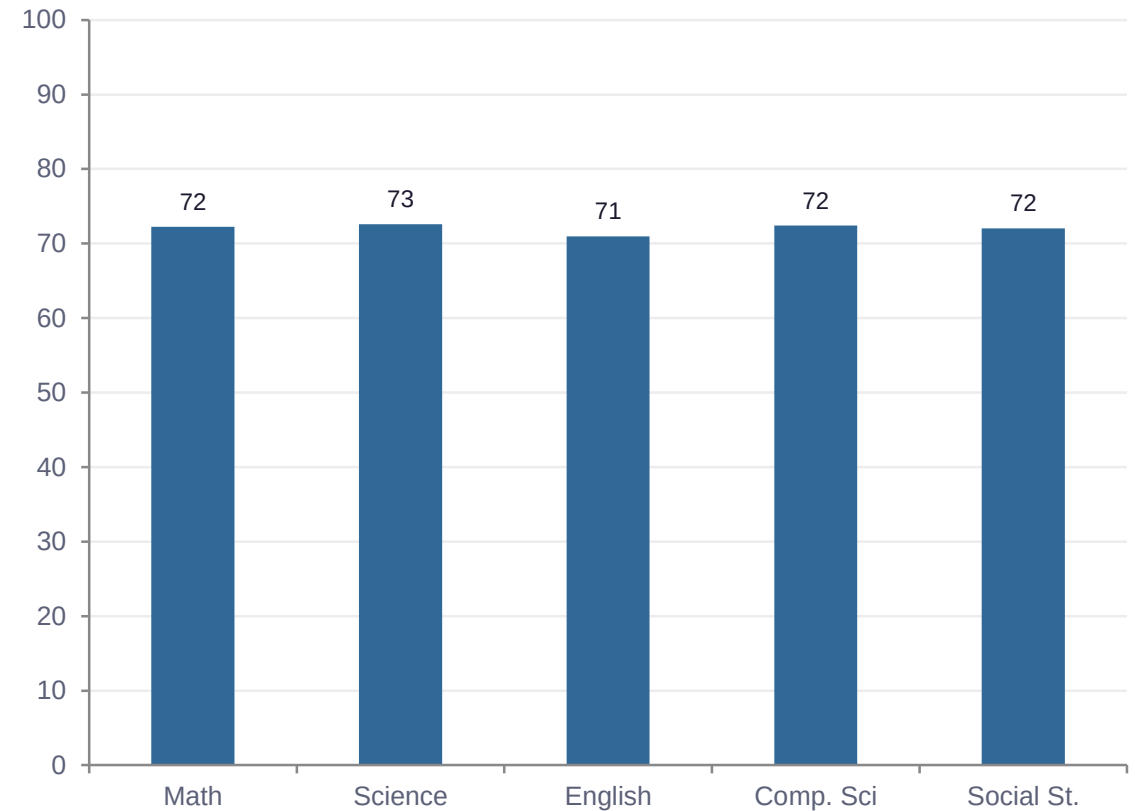
5

Subjects Covered

72.5

Avg Class Score

Class Average Score by Subject



PROJECT 2

Matrix Operations Tool

A menu-driven command-line application built with NumPy, offering a full set of matrix operations with input validation and clear error handling for invalid dimensions or singular matrices.

- Addition
- Subtraction
- Multiplication
- Scalar Multiplication
- Transpose
- Determinant
- Inverse



Built for Robustness

- ✓ Validates matrix dimensions before every operation
- ✓ Clear, human-readable error messages for mismatched dimensions
- ✓ Detects non-square matrices for determinant/inverse
- ✓ Detects singular matrices (zero determinant)
- ✓ Interactive menu — loops until the user exits

PROJECT 3

Linear Regression — House Price Prediction

Trained a scikit-learn Linear Regression model on 200 houses to predict price from area, bedrooms, bathrooms, age, and distance from the city centre — split 80/20 into training and test sets.

0.9966

R² Score

₹1.30L

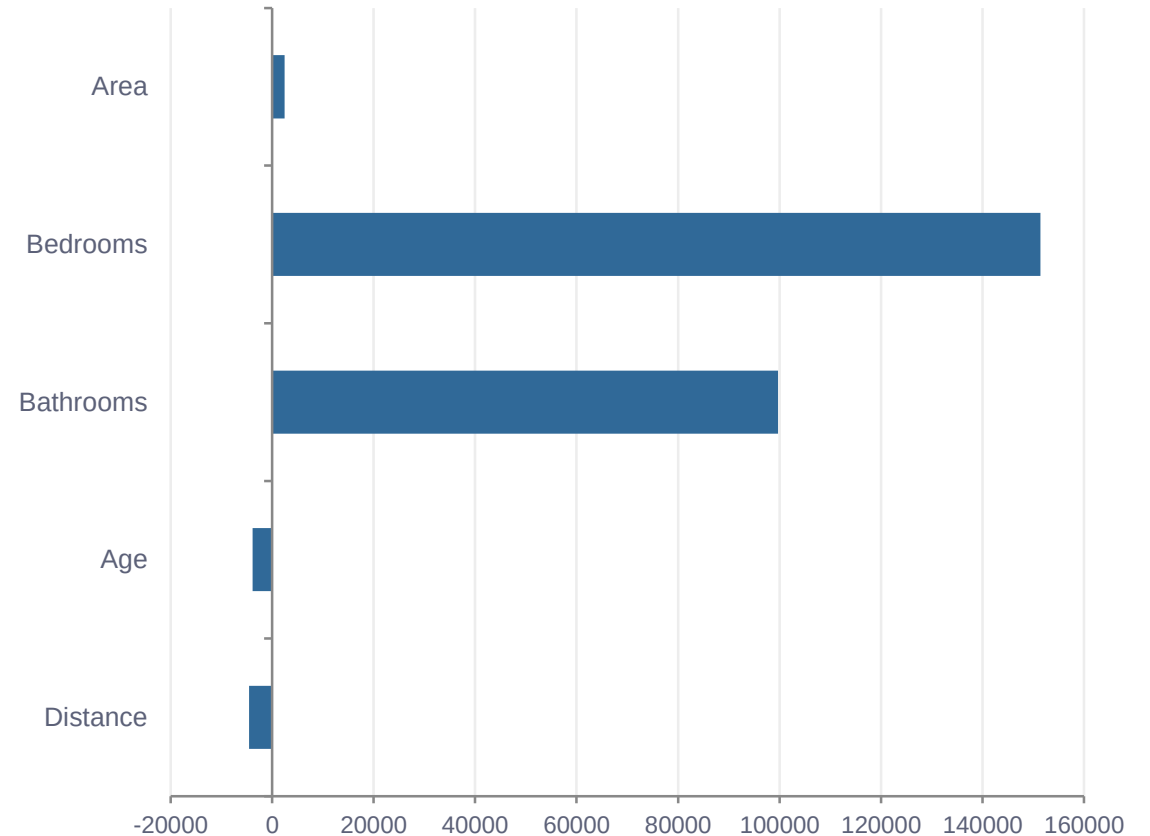
Mean Abs. Error

₹1.55L

RMSE

Bedrooms and Bathrooms have the strongest positive effect on predicted price; Age and Distance from city centre reduce it — consistent with real-world housing trends.

Feature Impact on Predicted Price



Tools & Technologies

The Python stack used consistently across all three projects



Python 3

Core programming language



pandas

Data loading, cleaning & analysis



NumPy

Numerical & matrix computation



matplotlib

Data visualization



scikit-learn

Machine learning model training



VS Code

Development environment



Git & GitHub

Version control & hosting

Outcomes & Key Learnings



3 complete Python projects

Data analysis, matrix computation, and machine learning — each independently built, tested, and documented.



Strong model performance

The regression model achieved an R^2 of 0.9966, demonstrating a reliable, well-validated predictive pipeline.



End-to-end workflow experience

From loading raw data to producing polished reports, charts, and a version-controlled GitHub repository.



Practical Python fluency

Hands-on practice with pandas, NumPy, matplotlib, and scikit-learn — the core data & ML toolkit used in industry.

Certificate of Completion

Issued by Squarcell Resource India Pvt. Ltd. (QSkill) on successful completion of the internship



Thank You

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